

# Dalton Kutzen

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## Education

**Brigham Young University — B.S. Molecular Biology**  
Provo, Utah

Jan 2025 – Apr 2028

## Experience

### Research Assistant

Sep 2025 – Present

Grose Lab, Brigham Young University · Provo, UT

- Prepared media, cultured bacteriophages with bacterial hosts, and conducted plaque and spot assays to evaluate phage activity and host interactions.
- Performed bioinformatic analyses to annotate and compare phage genomes, creating the **Phantom** Snakemake pipeline for reproducible annotation.
- Integrated wet lab microbiology techniques with computational genomics to investigate phage diversity, host range, and therapeutic applications.

### IT Service Desk Supervisor

May 2026 – Present

Brigham Young University — College of Life Sciences · Provo, UT

- Lead frontline IT support for faculty, staff, and students across the College of Life Sciences, resolving high-complexity hardware, software, and systems incidents.
- Mentor student technicians in troubleshooting best practices and train new hires on Life Sciences IT infrastructure, tools, and clear user communication.
- Drove internal automation, documentation, and reporting initiatives that reduced repetitive workflows and standardized support delivery college-wide.

### IT Service Desk Technician

Aug 2025 – May 2026

Brigham Young University · Provo, UT

- Oversaw deployment, security, and maintenance of 1,000+ campus workstations and lab computers, minimizing system downtime.
- Supported a community of 5,000+ users by troubleshooting hardware, software, and wireless networking issues for academic and administrative operations.

## Selected Projects

### Phantom — Bacteriophage Analysis Workflow

Jun 2026 – Present

- Modular Snakemake pipeline for phage genome annotation and downstream comparative analysis.
- Uses isolated Conda environments and declarative workflow rules to make runs reproducible across systems.
- Produces organized intermediate files, results, logs, and publication-oriented visualizations.

## Technical Skills

| Area                        | Tools & Proficiencies                                                                                                   |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Computational Biology       | Comparative genomics, Snakemake, Conda, Biopython                                                                       |
| Wet Lab & Molecular Biology | Phage and bacterial culturing, plaque & spot assays, host range testing, media preparation, agarose gel electrophoresis |
| Programming & Data          | Python, R, Bash, SQL, Git, pandas, tidyverse, ggplot2, Quarto                                                           |
| IT & Systems Infrastructure | IT Service Management (ITSM), Linux server administration, Proxmox, Docker, 1,000+ endpoint management, networking      |
| Operations & Leadership     | project management, technical mentoring, documentation                                                                  |